

Introduction

With the increasing modern focus on renewable energy infrastructure, understanding how wind turbine blades degrade over time is an important step to their maintenance and optimization. Wind turbine blades exhibit trends of leading edge erosion (LEE) where rain and hail physically impact the external turbine coating, affecting its aerodynamics. The focus of my research this semester was to study and code different mathematical functions that estimate the distribution of raindrop sizes during various weather events. The overall goal was to incorporate these functions into an overall algorithm that estimates the lifecycle of a turbine with recorded weather data.

Method

I primarily used data from government Automated Surface Observing Systems (ASOS) which, among other things, provided measurements of rainfall rates (mm/hr) at given moments in time. I then input rainfall rates into two functions: the first alternative function I used came from J. S. Marshall and W. McK. Palmer's paper and the second I developed from Dolan et. al.'s paper, outputting the estimated number of drops per possible droplet diameter. In other words, for a range of theoretical diameters, the functions estimate the number of raindrops that could feasibly exist given that rainfall rate.

Conclusion

Using a general script provided by Professor Pryor, I incorporated the droplet distribution functions to estimate the lifecycle of wind turbines and compare the differences in estimates across the different functions. I conclude that there is significant agreement between the different functions on the moments in which the greatest turbine damage occurs. This is important because methods of combining the algorithms and finding agreement across their data output could be very effective in predicting and identifying, with confidence, concrete moments to slow blade speed or even shut a turbine down completely, extending the duration that wind turbines can be used to generate power.

Works Cited

- Dolan, B., B. Fuchs, S. A. Rutledge, E. A. Barnes, and E. J. Thompson. "Primary Modes of Global Drop Size Distributions". *Journal of the Atmospheric Sciences* 75.5 (2018): 1453-1476. <https://doi.org/10.1175/JAS-D-17-0242.1>
- Marshall, J. S. , and W. McK. Palmer. "THE DISTRIBUTION OF RAINDROPS WITH SIZE". *Journal of Atmospheric Sciences* 5.4 (1948): 165-166. [https://doi.org/10.1175/1520-0469\(1948\)005<0165:TDORWS>2.0.CO;2](https://doi.org/10.1175/1520-0469(1948)005<0165:TDORWS>2.0.CO;2)